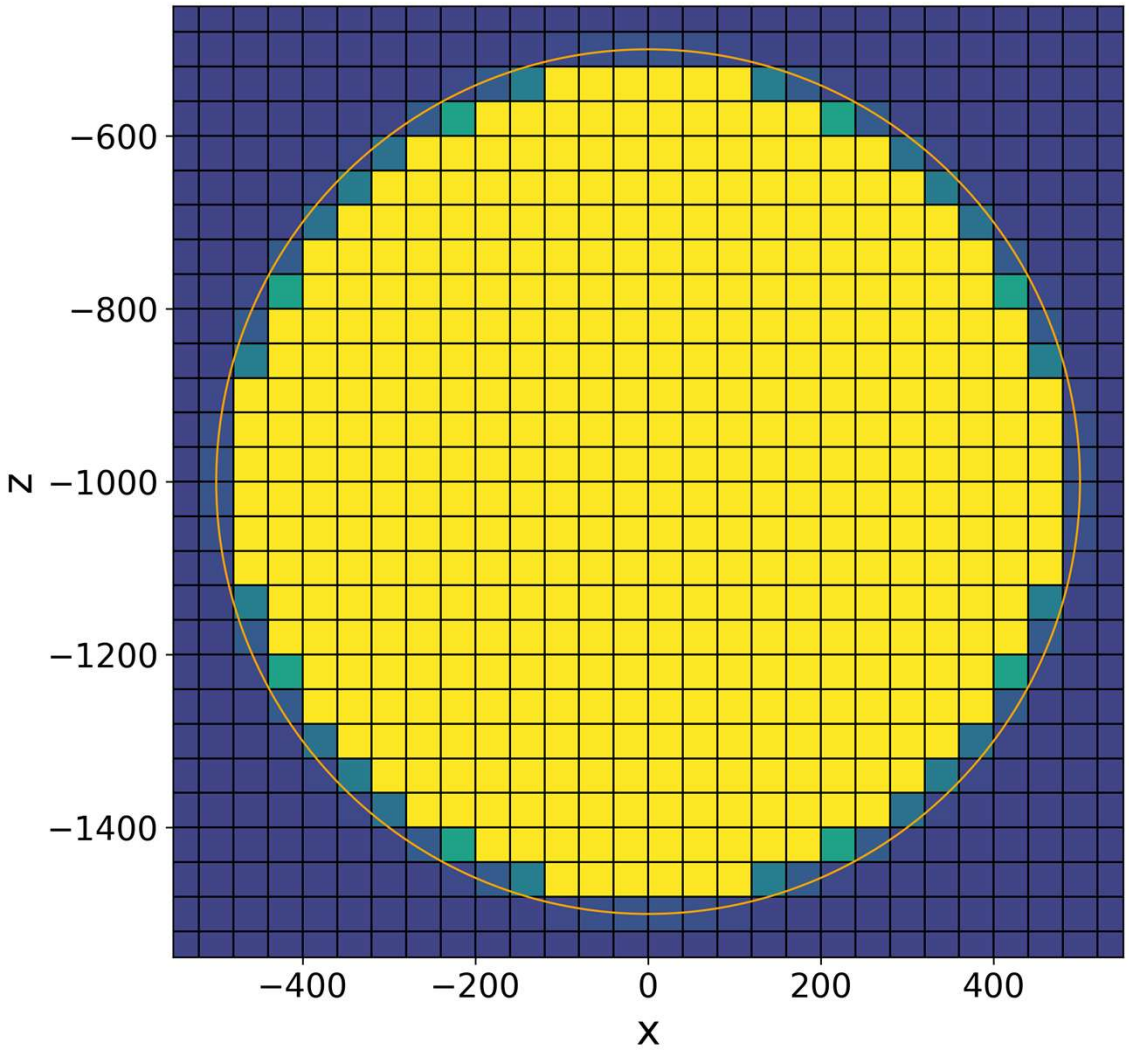
Techniques for Improving MT Forward Simulations with Octree Meshes

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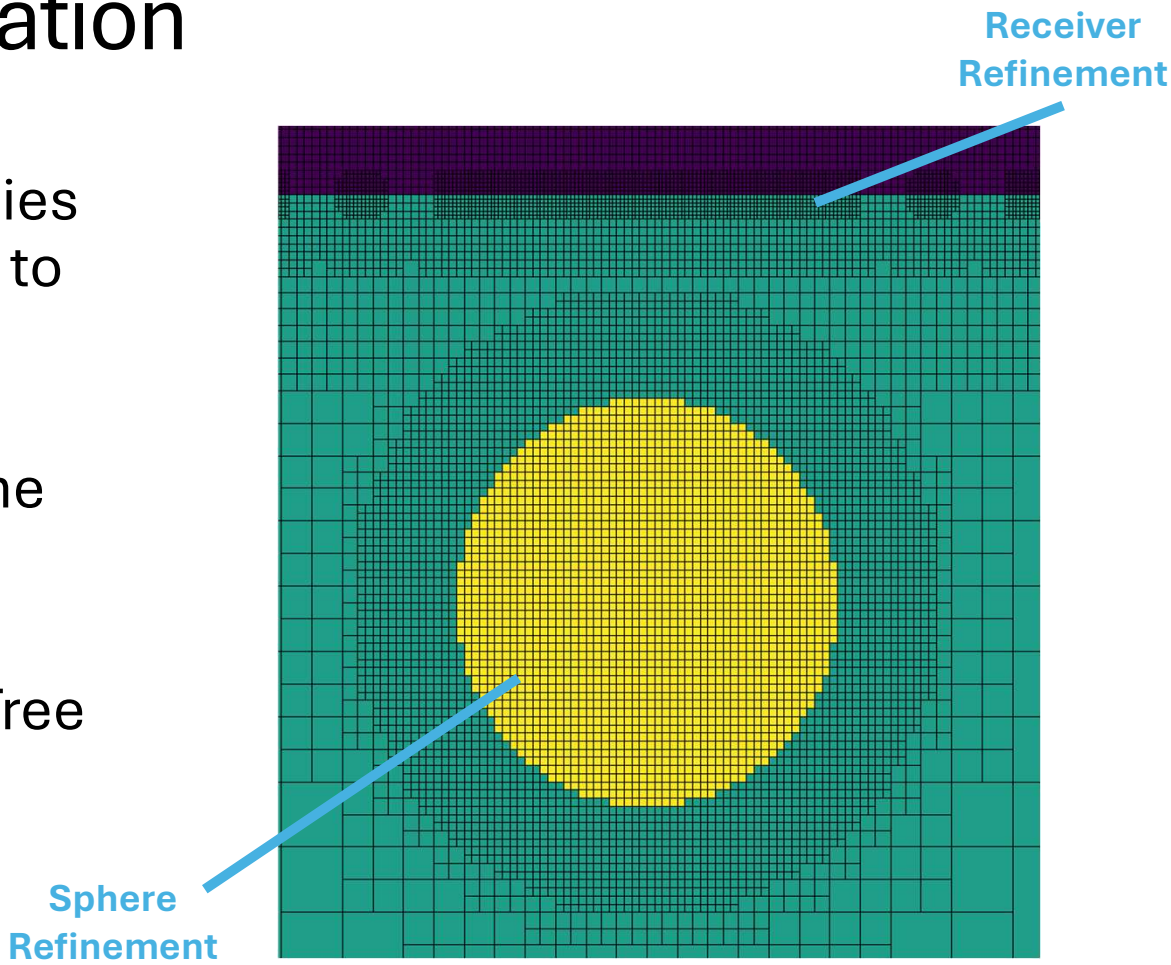


Motivation and Abstract

- The sphere in a halfspace example was used to benchmark natural source forward simulations using the SimPEG framework
- Tests were conducted with the eventual goal of inversion in mind, and different OcTree mesh design strategies were studied to:
    1. Minimize computation time
    2. Maximize simulation accuracy
- Volume averaging was also explored as a method of improving accuracy without increasing computation time

Localized Discretization

- We are looking at frequencies from 1mHz ($\delta \approx 500$km) up to 10kHz ($\delta \approx 150$m)
- The mesh needs to be discretized finely around the receivers and the bodies being imaged
- How can we refine our OcTree mesh in the most efficient way possible?



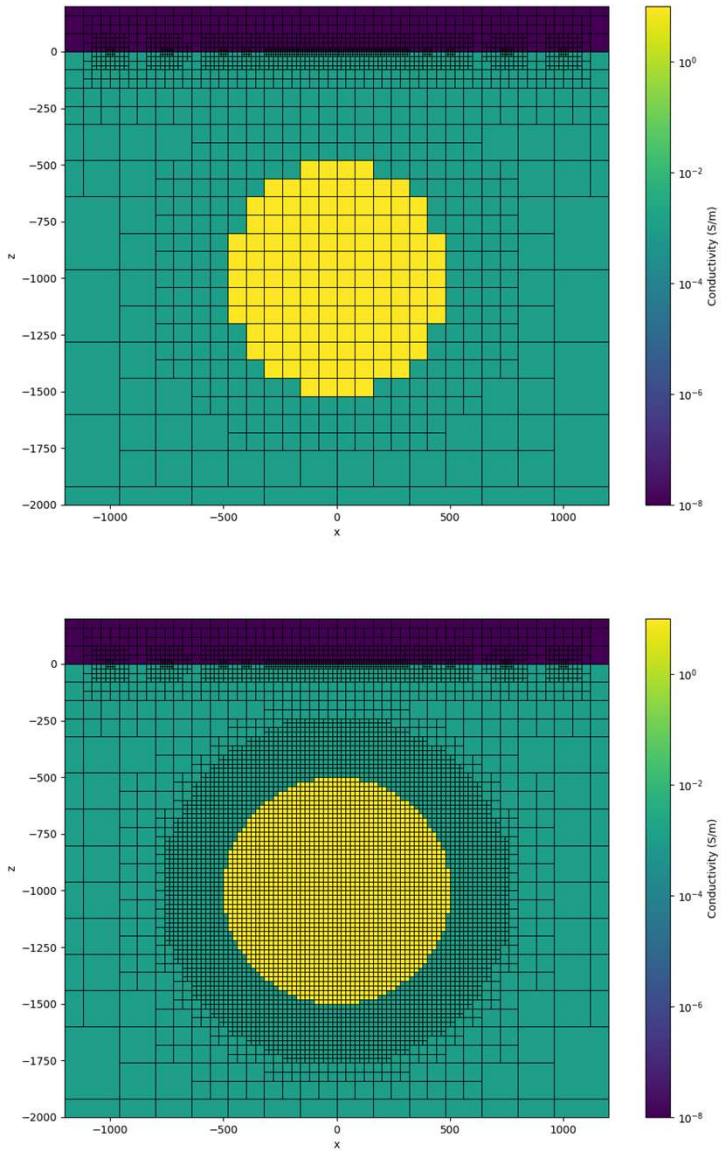
Adjusting the Sphere

- Receiver refinement is fixed at 10m cells while sphere refinement is varied
- This should mainly affect the results of lower frequencies near the center of the profiles

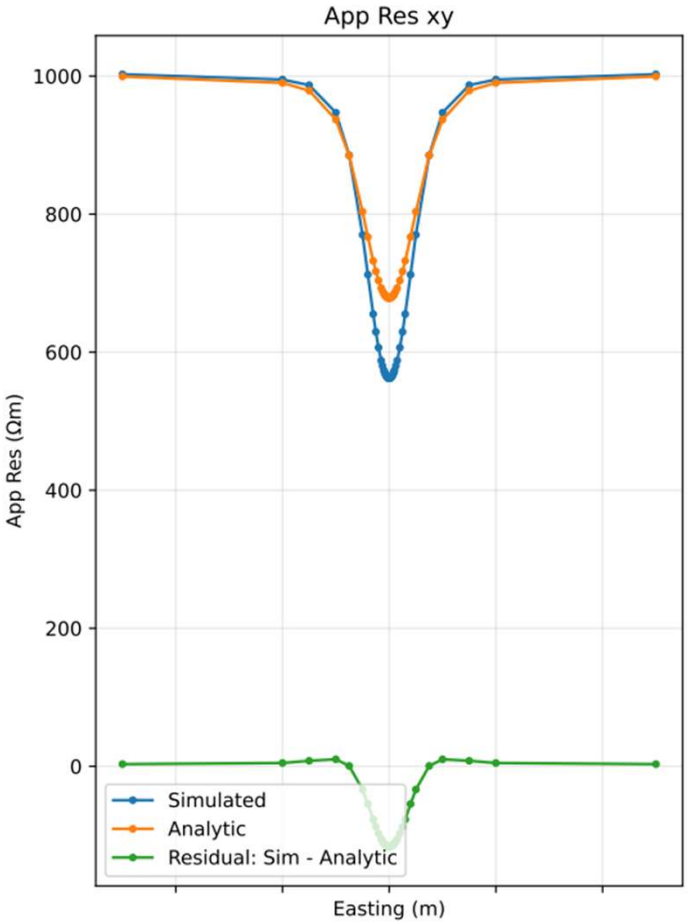
80m

↓

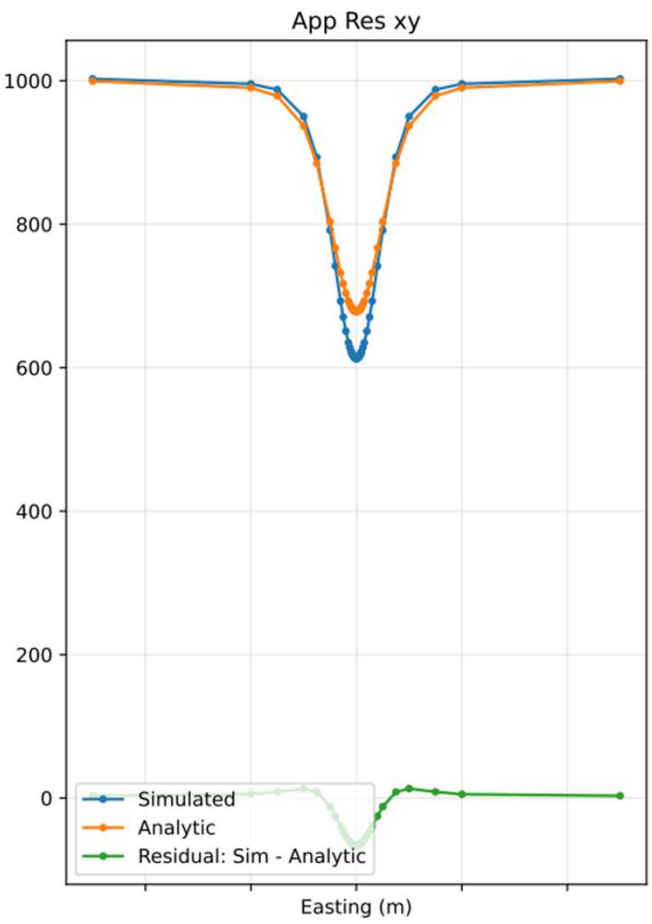
20m

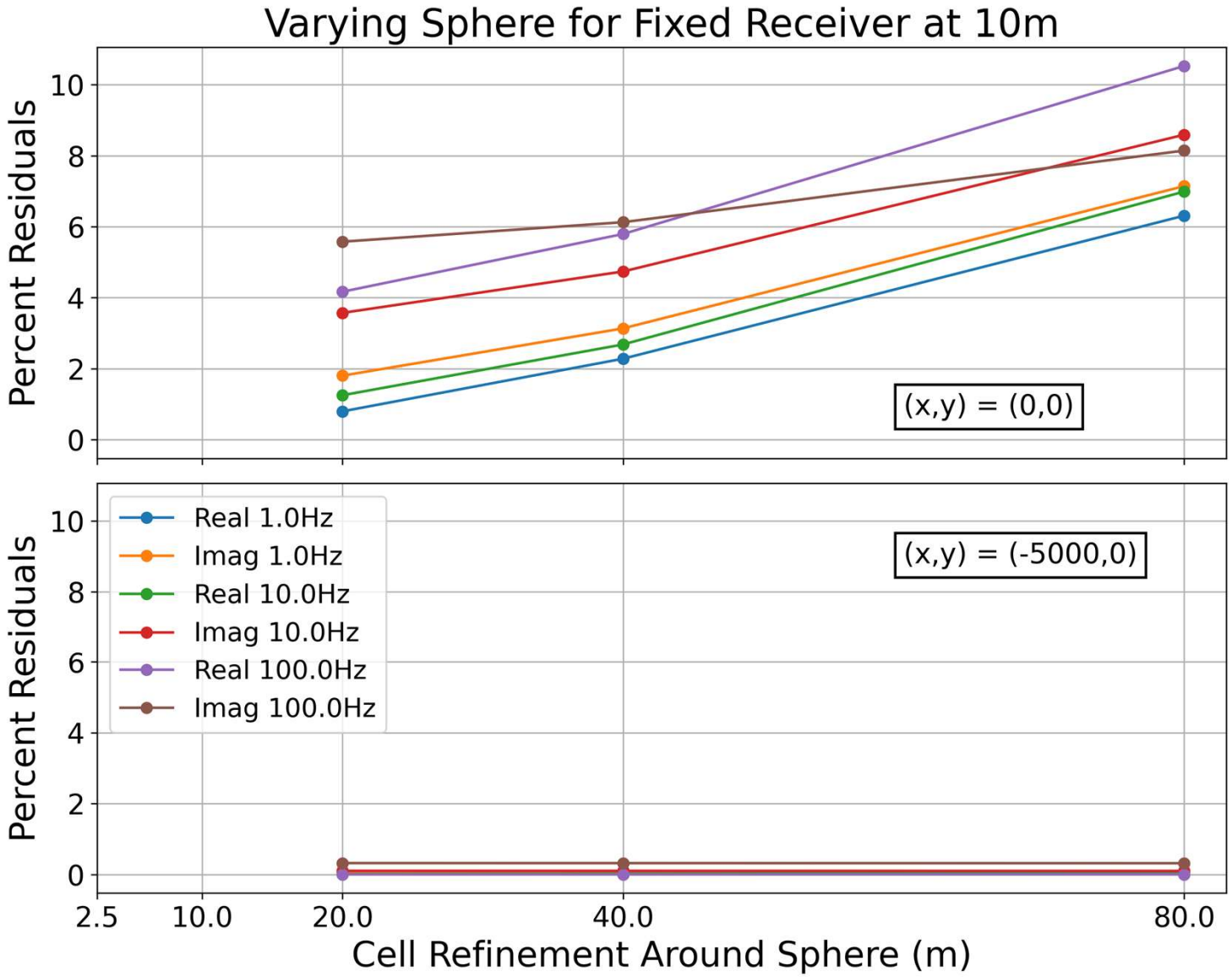


Apparent Resistivity 100Hz



80m → 20m




Varying Sphere for Fixed Receiver at 10m
Percent Residuals
(x,y) = (0,0)
(x,y) = (-5000,0)
Real 1.0Hz
Imag 1.0Hz
Real 10.0Hz
Imag 10.0Hz
Real 100.0Hz
Imag 100.0Hz
0
2
4
6
8
10
2.5
10.0
20.0
40.0
80.0
Cell Refinement Around Sphere (m)

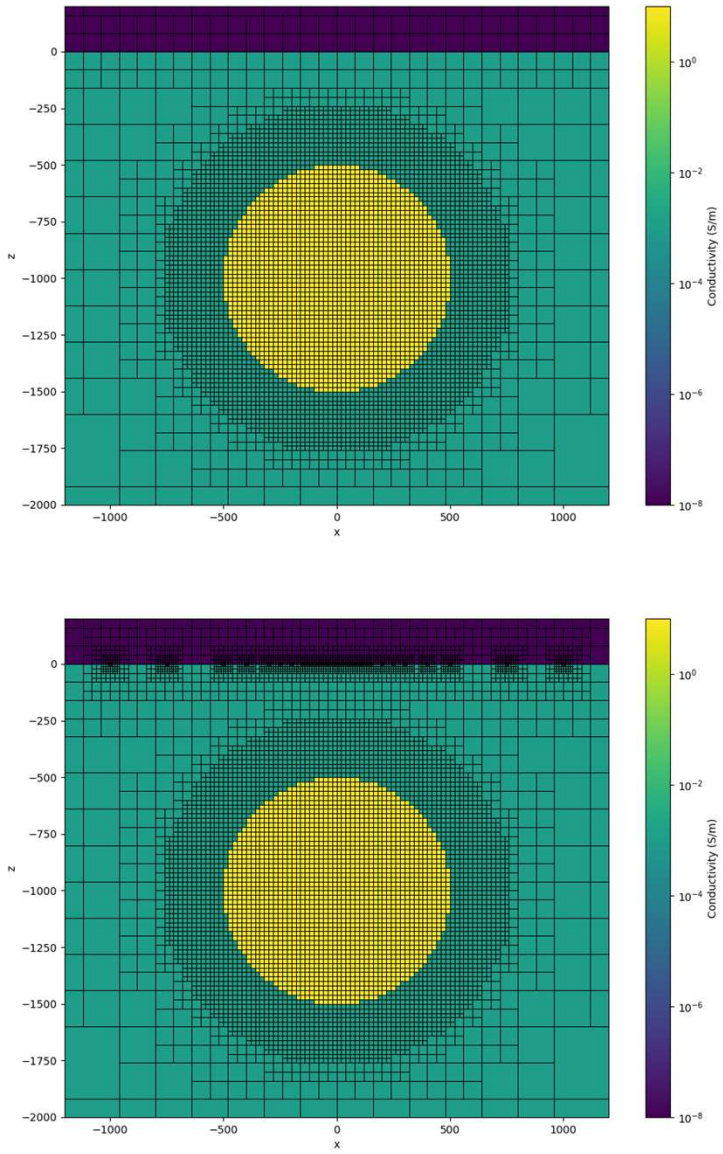
Adjusting the Receivers

- Sphere refinement is fixed at 20m cells while receiver refinement is varied
- This should mainly affect the results of higher frequencies far from the center of the profiles

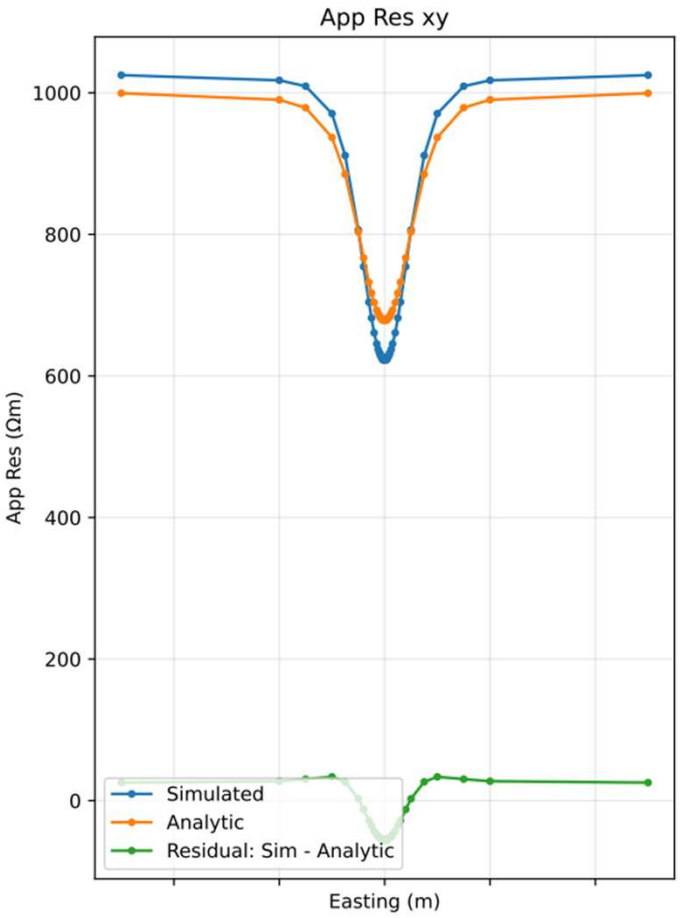
80m

↓

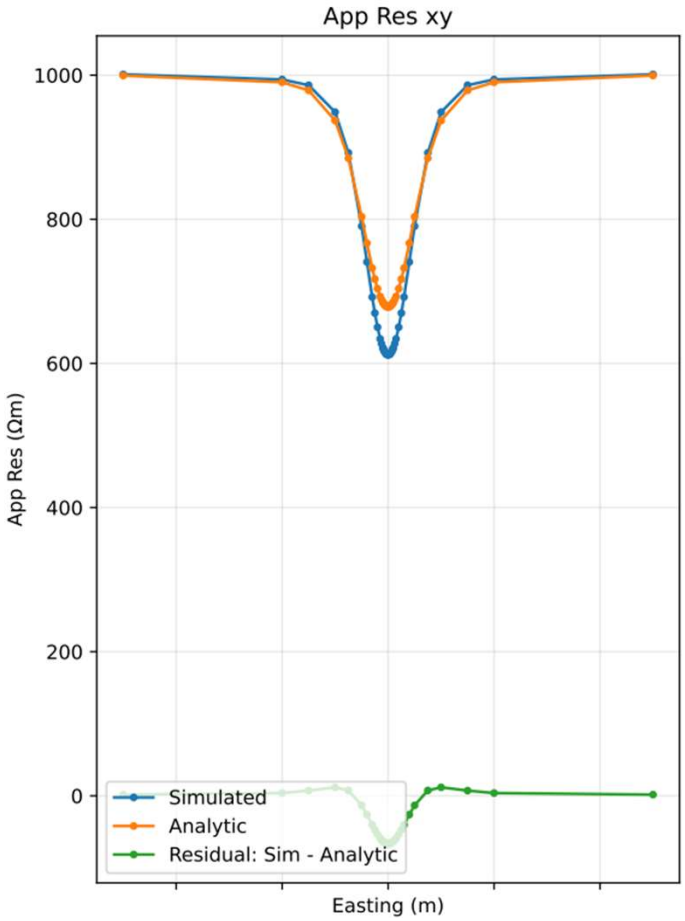
5m

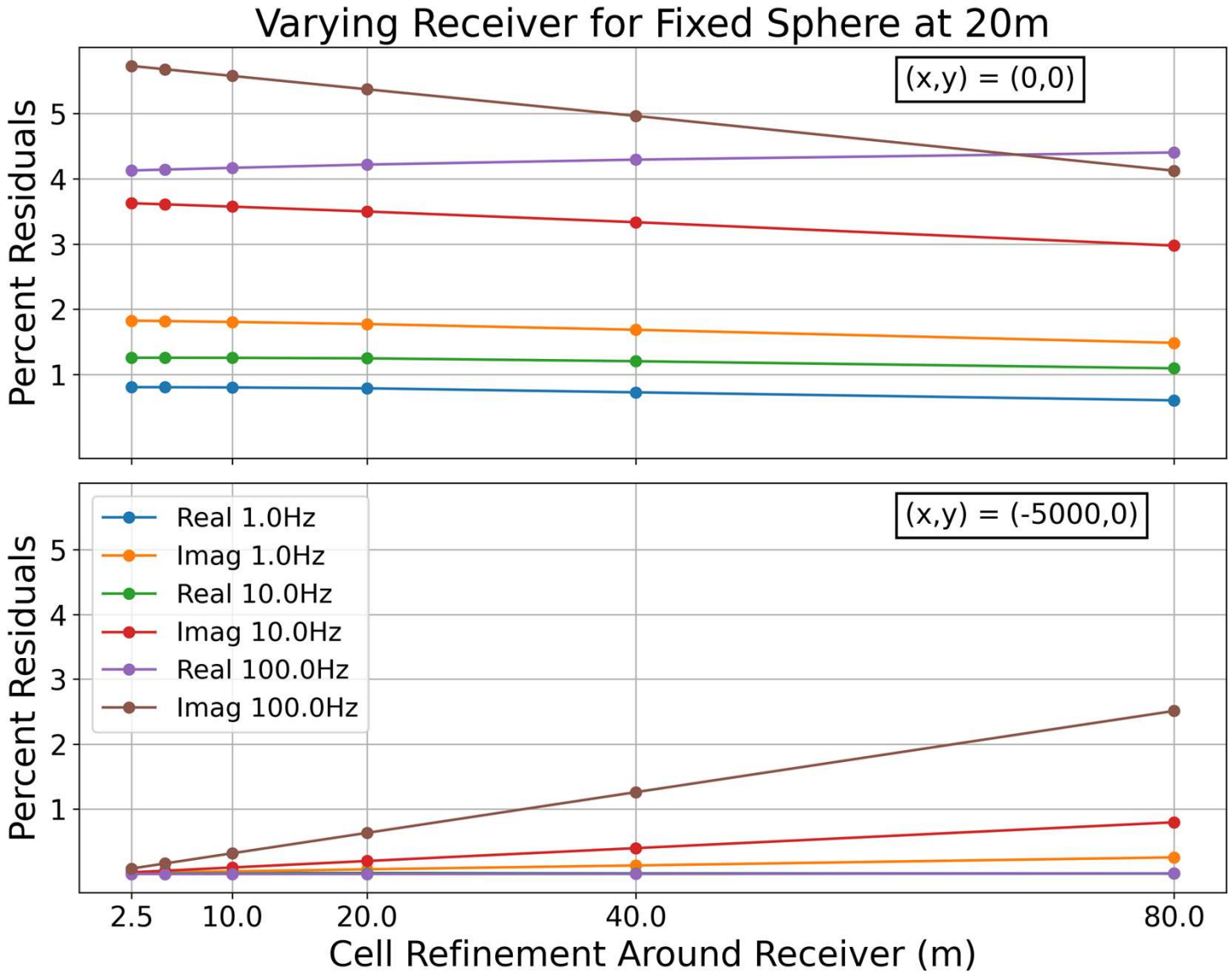


Apparent Resistivity 100Hz



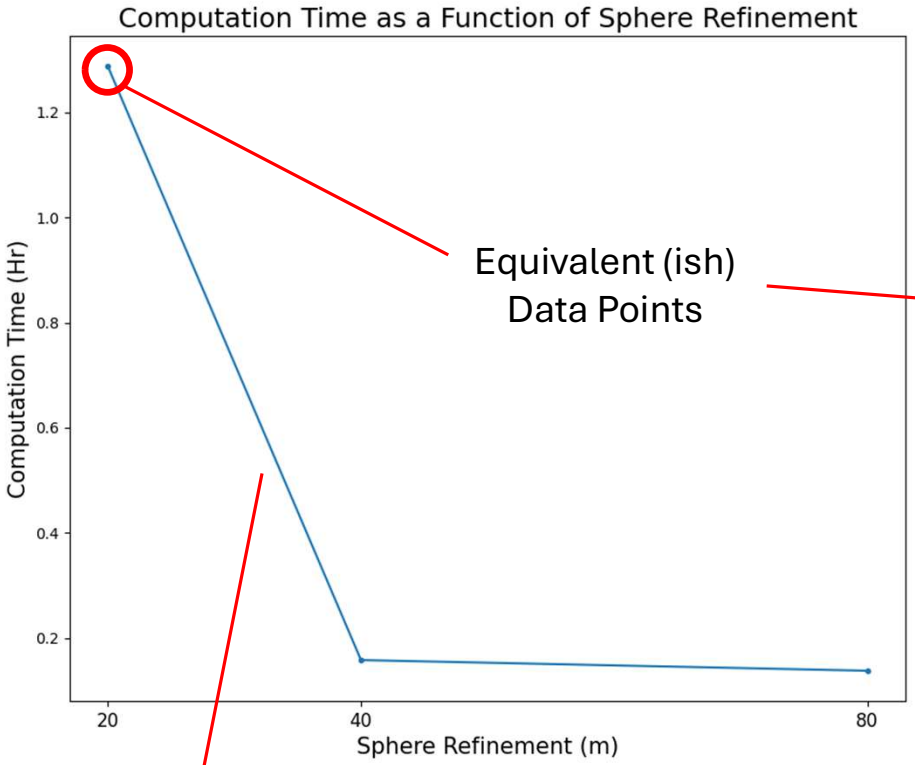
80m → 20m

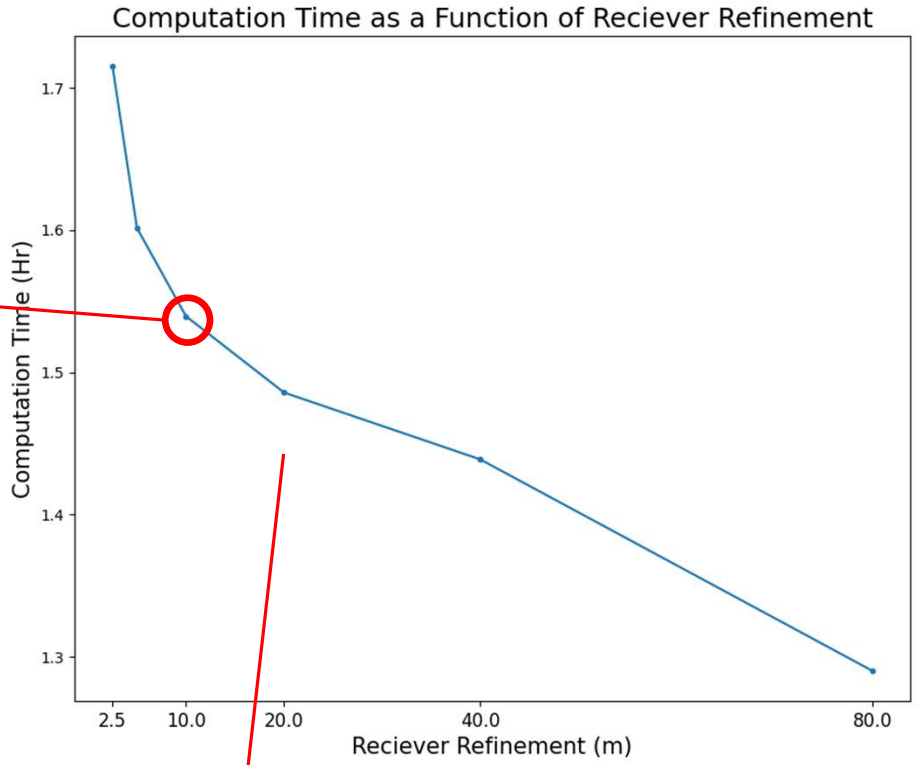



Varying Receiver for Fixed Sphere at 20m
(x,y) = (0,0)
(x,y) = (-5000,0)
Real 1.0Hz
Imag 1.0Hz
Real 10.0Hz
Imag 10.0Hz
Real 100.0Hz
Imag 100.0Hz
Percent Residuals
Percent Residuals
1
2
3
4
5
2.5
10.0
20.0
40.0
80.0
Cell Refinement Around Receiver (m)

How do these parameters influence computation time?

Computation for 8 Frequencies in Serial





Equivalent (ish)
Data Points

Step Dropoff (~x8)

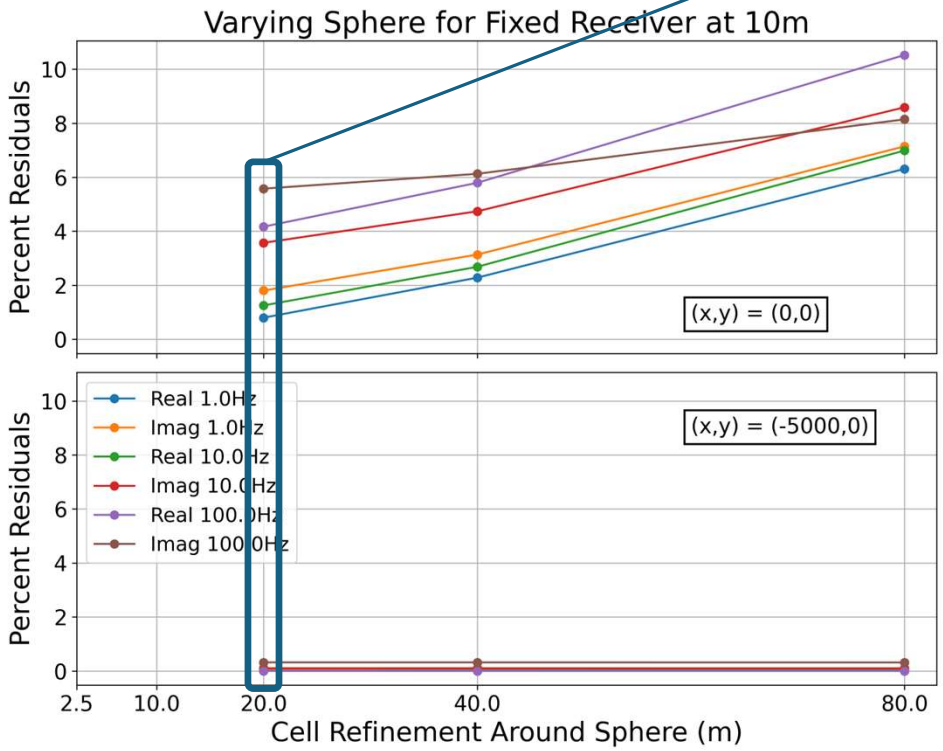
Linear Dropoff?

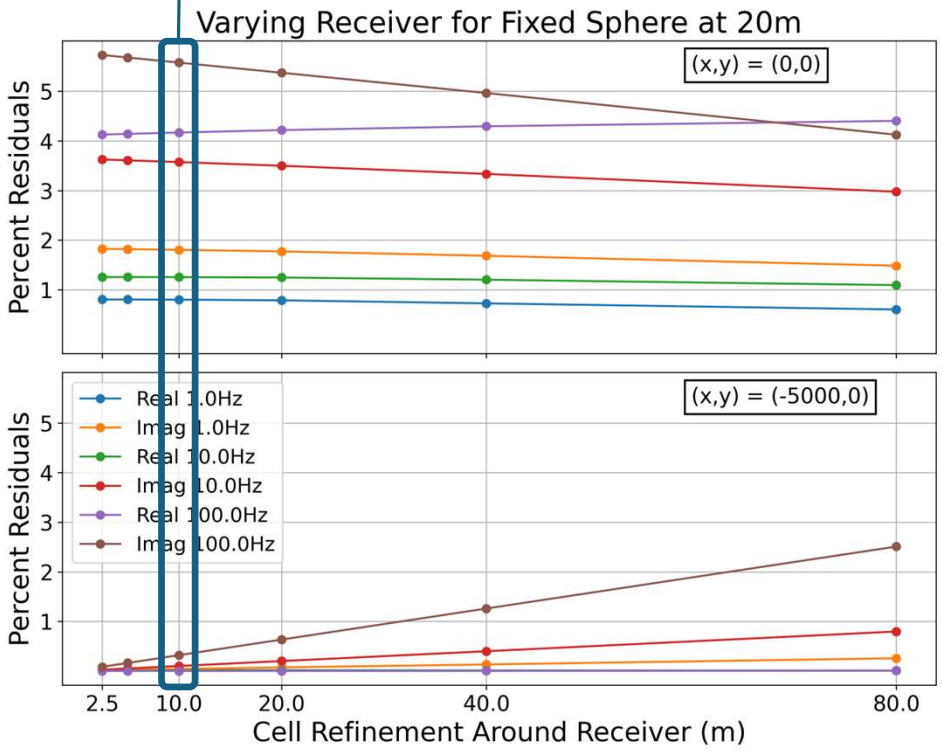
Current Theory

- The cells added via receiver discretization are background cells
- This means that in SimPEG, they do not contribute to the size of the right-hand side of the solved equation (since only cells varying from background are included here)
- This may explain the receiver cells have less impact on computation time (we haven't fully tested this)
- This means this effect would more-or-less disappear in an inversion following the first iteration ☹

Tradeoff Takeaways

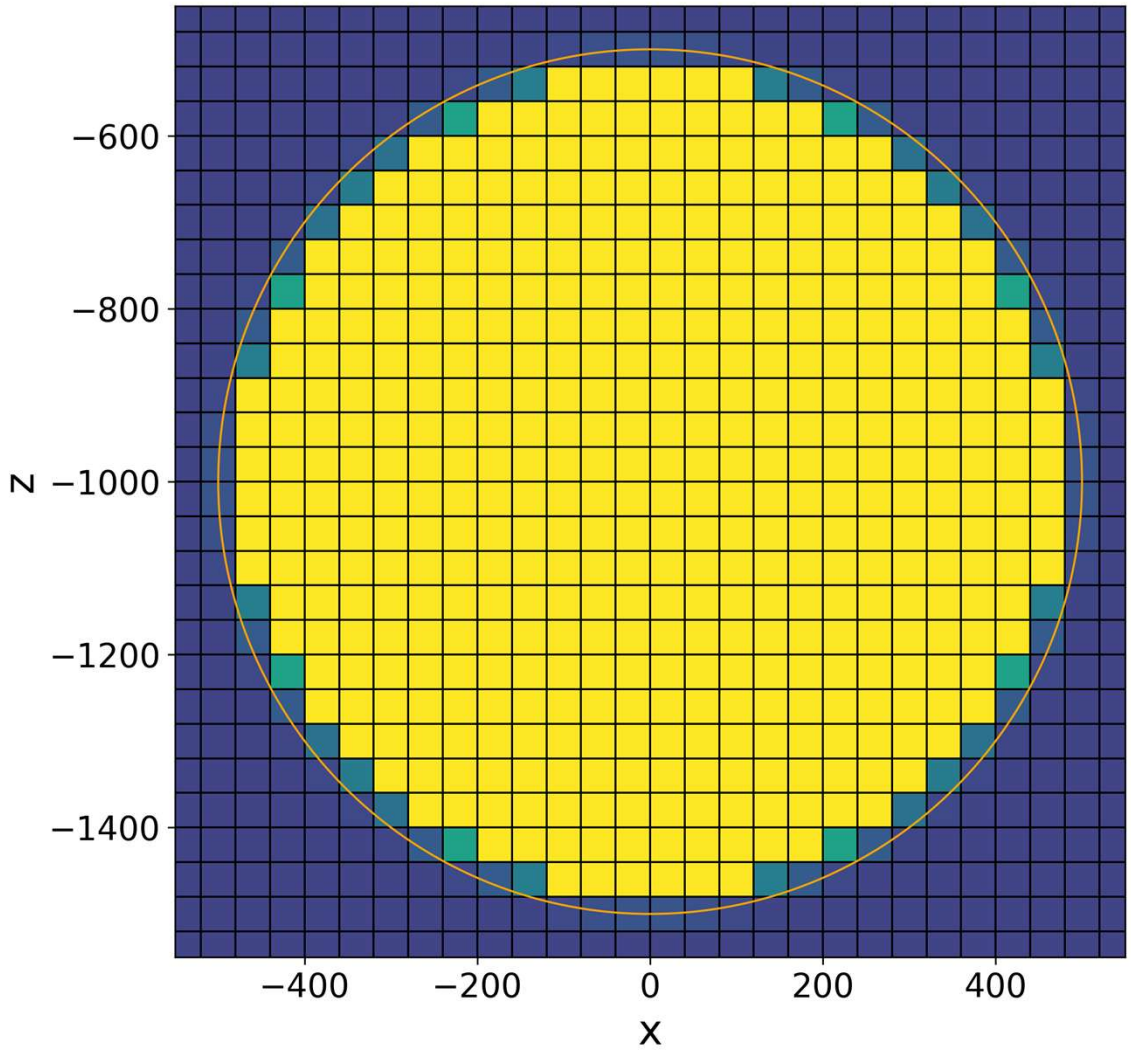
Equivalent and within 5% (almost) → 40m sphere and 5m receiver





Volume Averaging

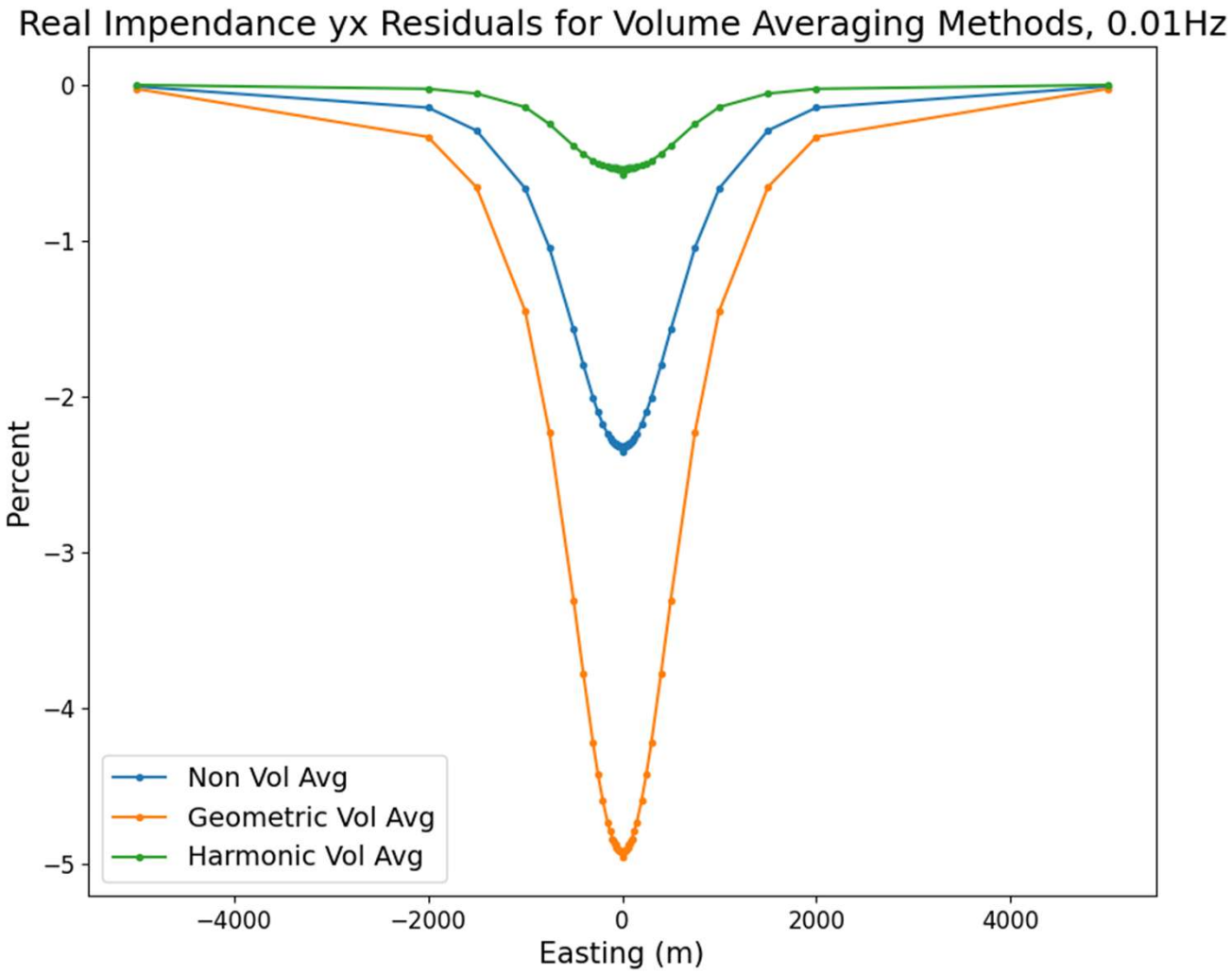
- Cells on the boundary of conductivity changes are assigned a value based on the fraction of the cell volume on each side of the boundary
- Can improve accuracy without requiring more cells

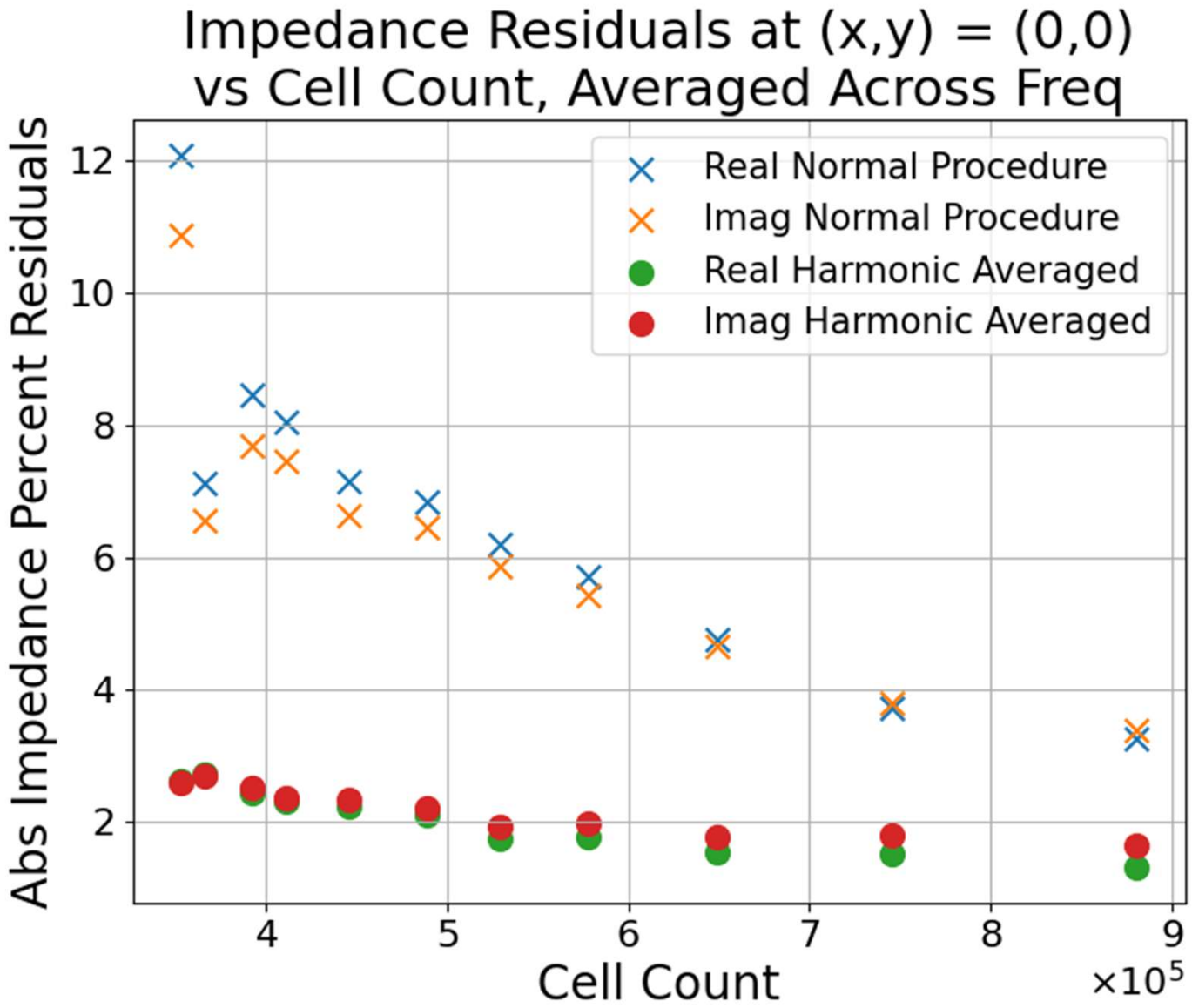


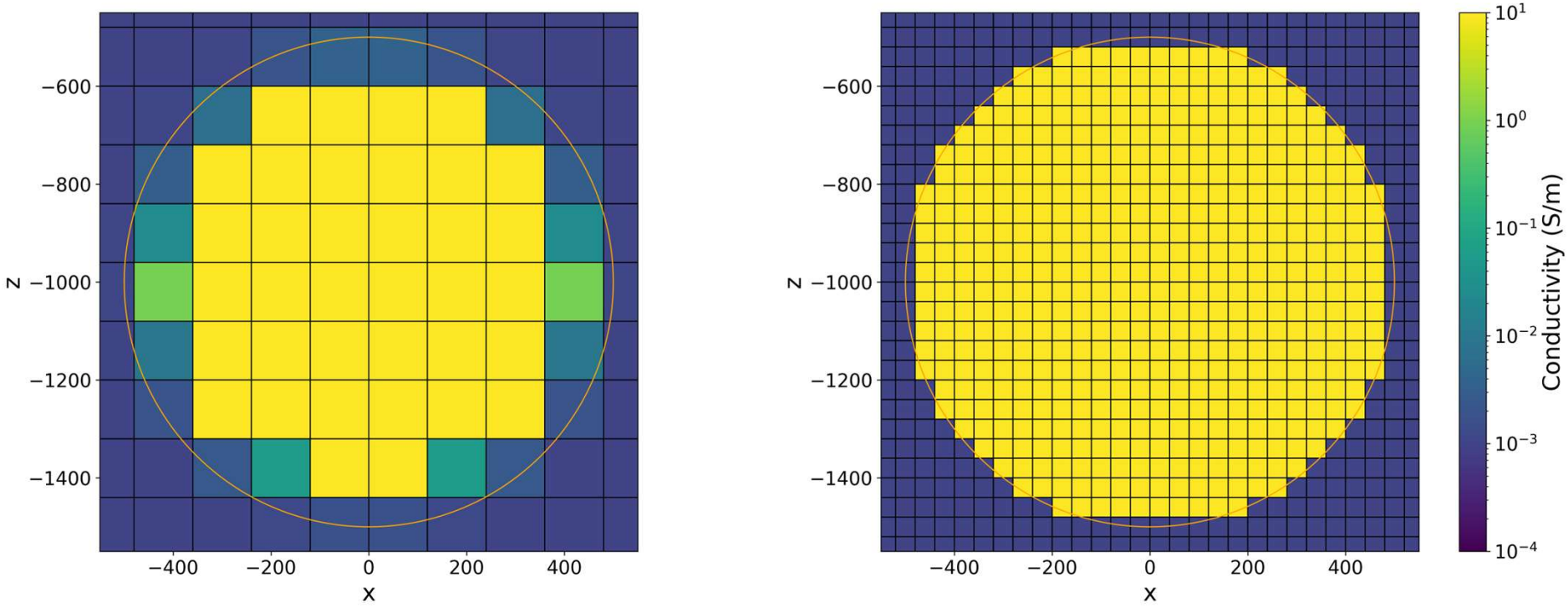
Volume Averaging Techniques

- Two main techniques tested:
  - Harmonic averaging: $\sigma = (\sum f_i {\sigma_i}^{-1})^{-1} \rightarrow \sigma = \frac{1}{f_1 {\sigma_1}^{-1} + f_2 {\sigma_2}^{-1}}$
  - Geometric averaging: $\sigma = \prod {\sigma_i}^{f_i} \rightarrow f = {\sigma_1}^{f_1} {\sigma_2}^{f_2}$
- The behavior of other averaging techniques can generally be captured by studying the above two since harmonic skews towards the lower conductivity and geometric skews towards the higher conductivity

- The trend shown is consistent for other frequencies (harmonic better)
- This implies we want to favour schemes that average towards background more
- Harmonic can often half percent residuals from standard cutoff



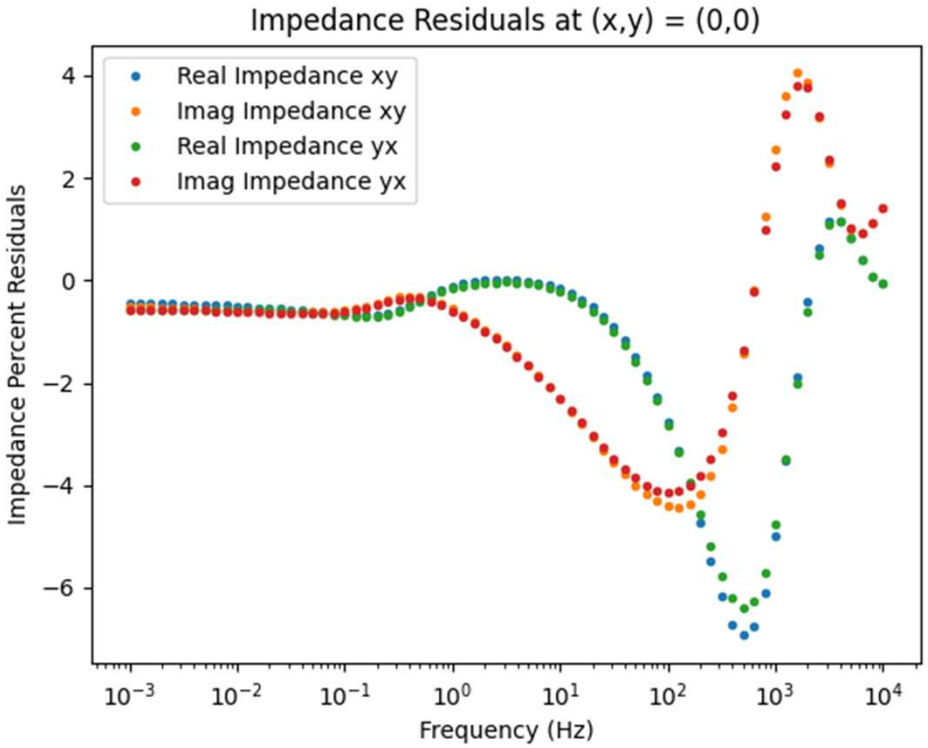

Impedance Residuals at (x,y) = (0,0)
vs Cell Count, Averaged Across Freq
Abs Impedance Percent Residuals
Cell Count
×10^5
Real Normal Procedure
Imag Normal Procedure
Real Harmonic Averaged
Imag Harmonic Averaged
2
4
6
8
10
12
4
5
6
7
8
9

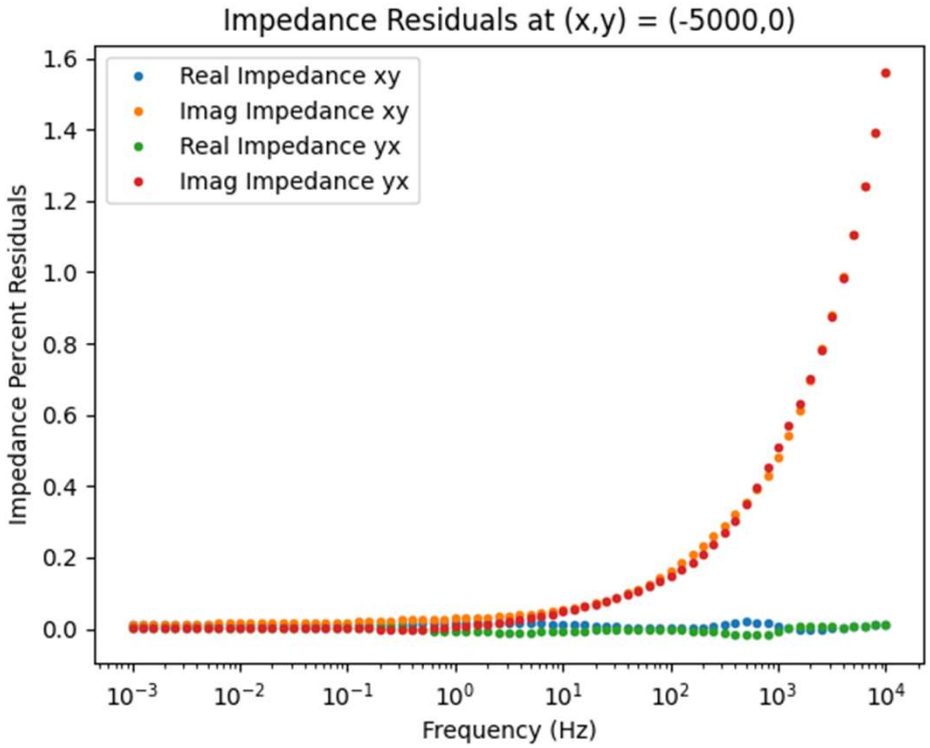

−600
−800
−1000
−1200
−1400
z
−400
−200
0
200
400
x
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
Conductivity (S/m)

Looking at our Submitted Model

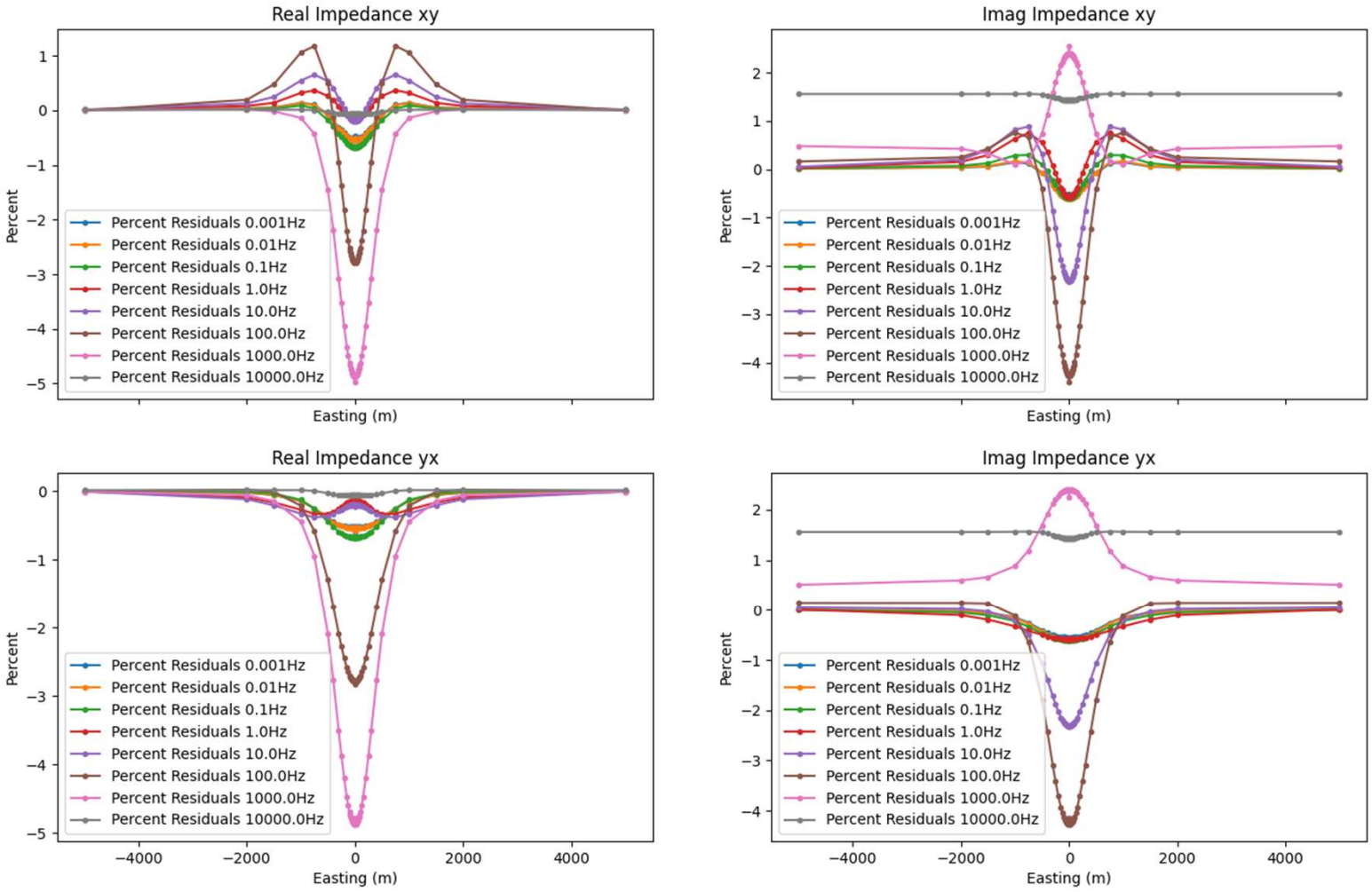
- 40m sphere cells, 5m receiver cells, 880,944 total cells, harmonic volume averaged
- Nominal cell size in core area 640m
- Computation time: 4.66 hours for 71 frequencies in serial
- 3.9 GHz Intel 6972P CPU, Nibi Cluster
- Peak memory allocation of 31.16 Gb

Percent Residuals per Frequency

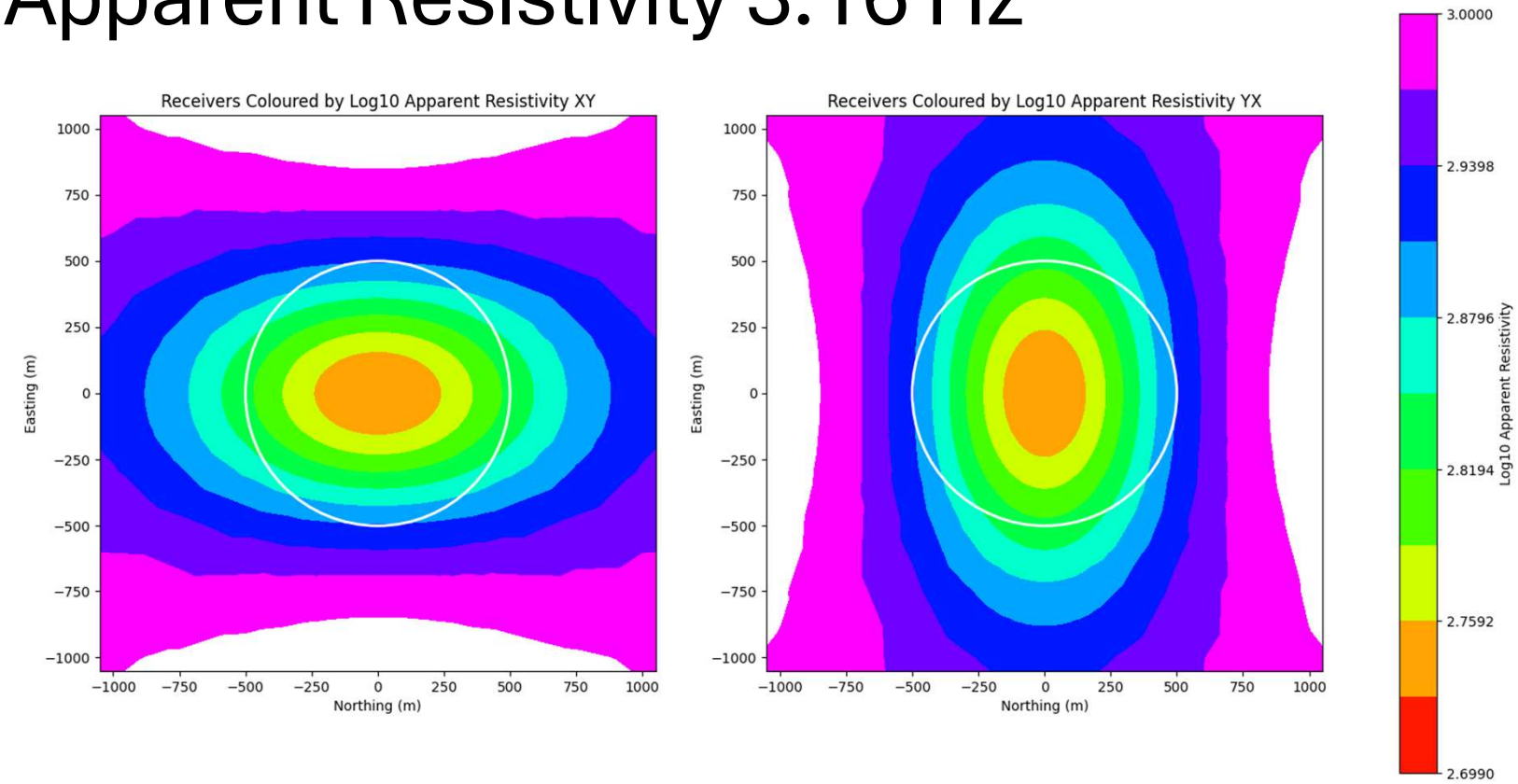




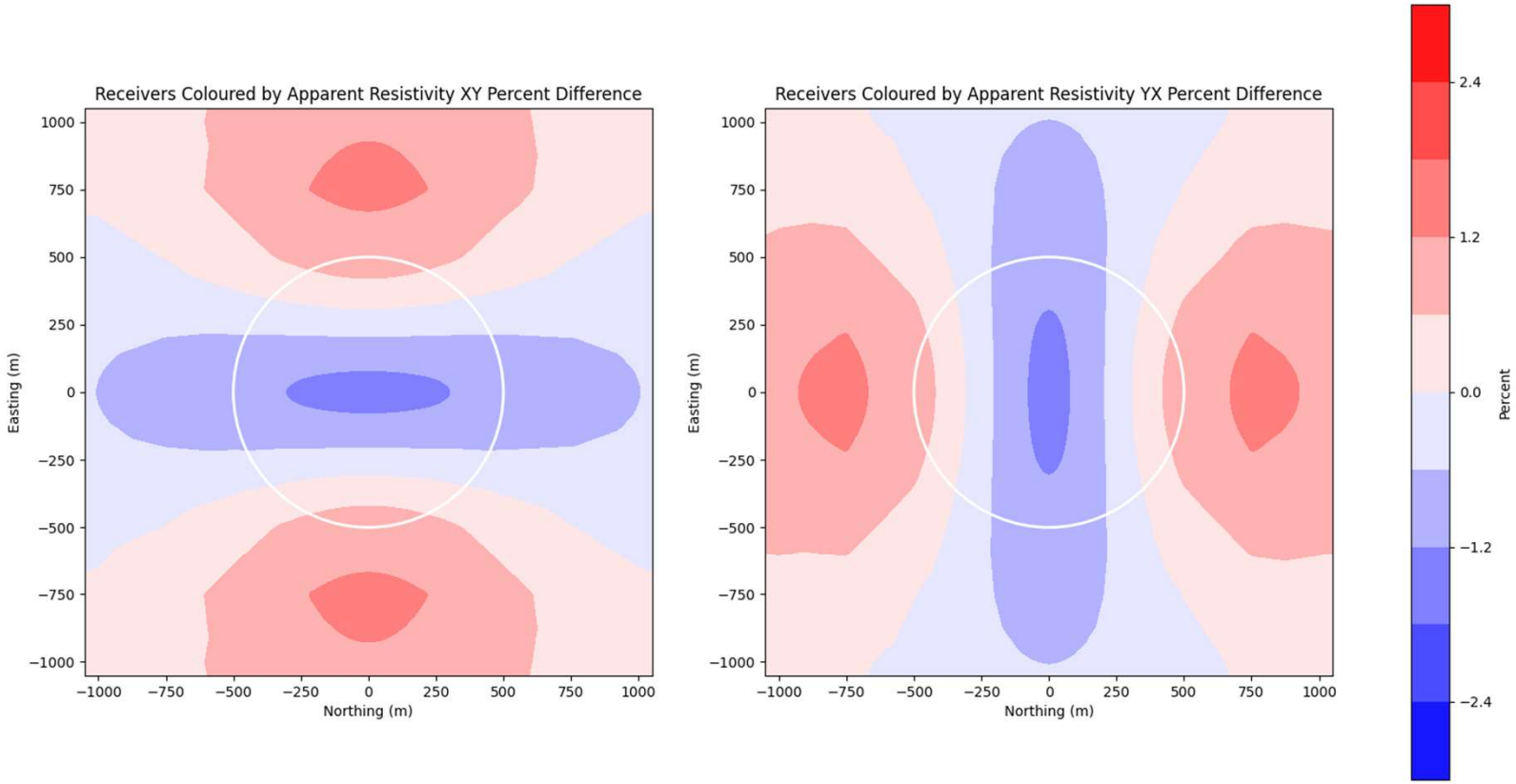
Impedance Residuals along Cut at y=0



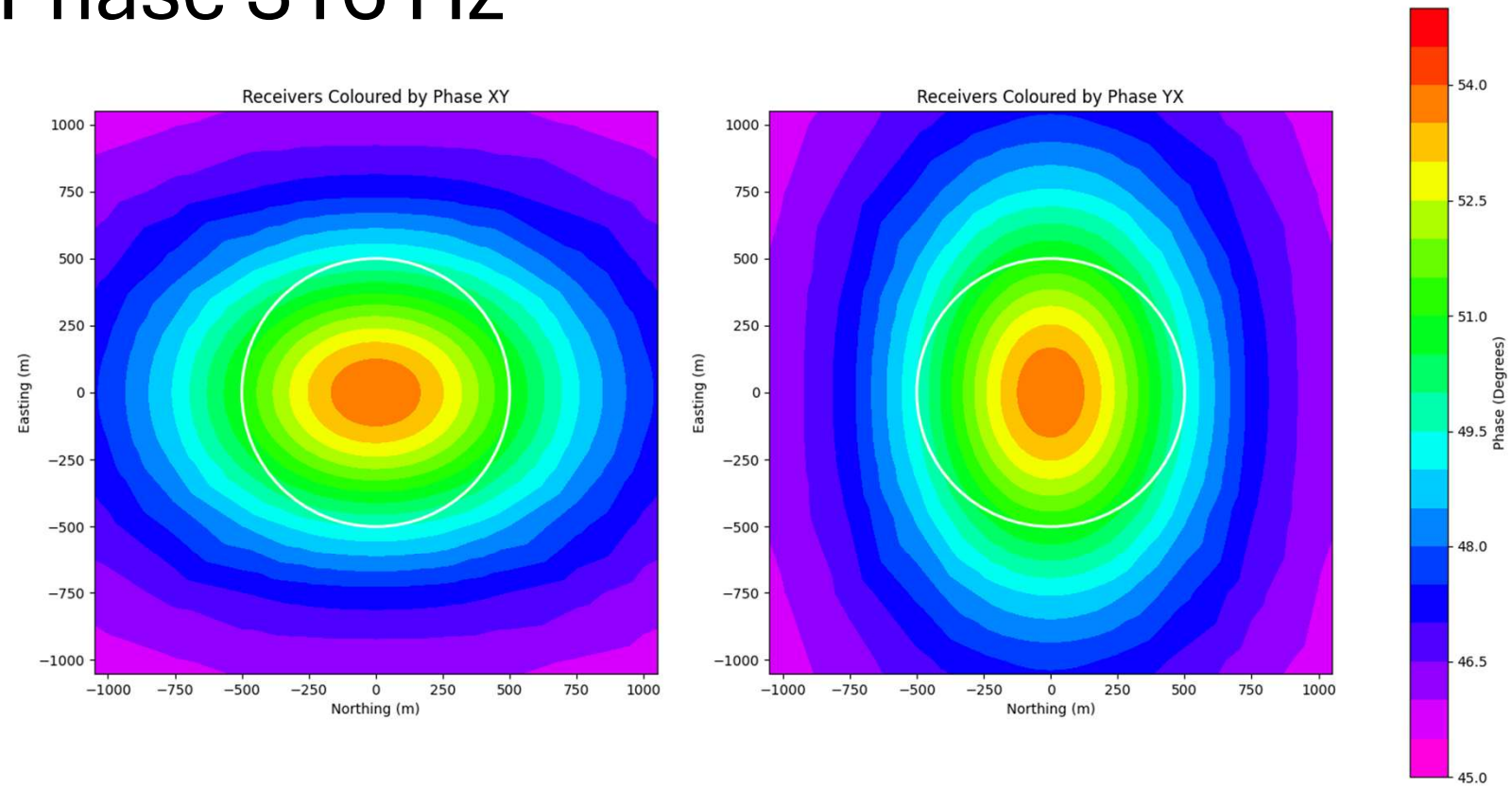
Apparent Resistivity 3.16 Hz



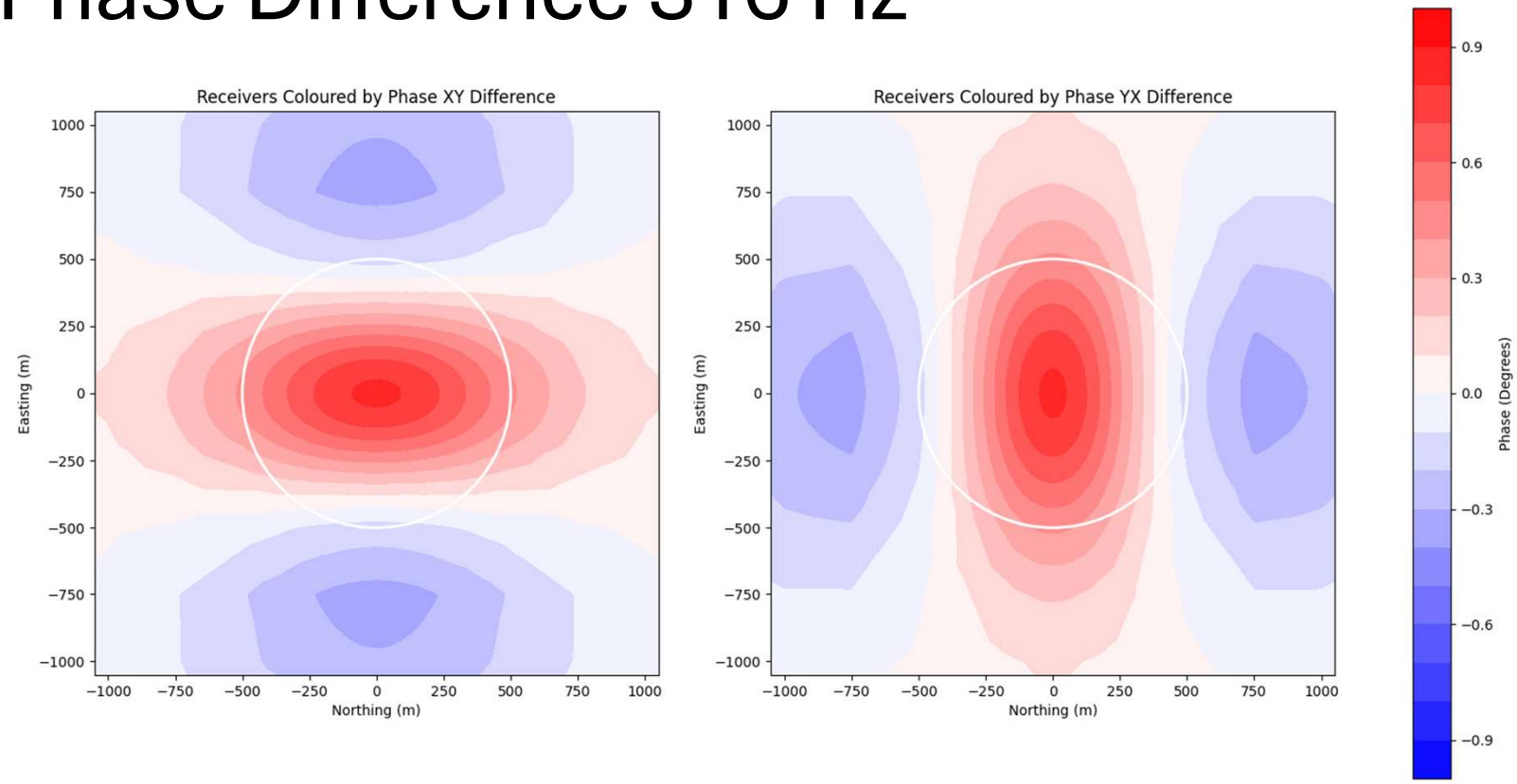
Apparent Resistivity Difference 3.16 Hz



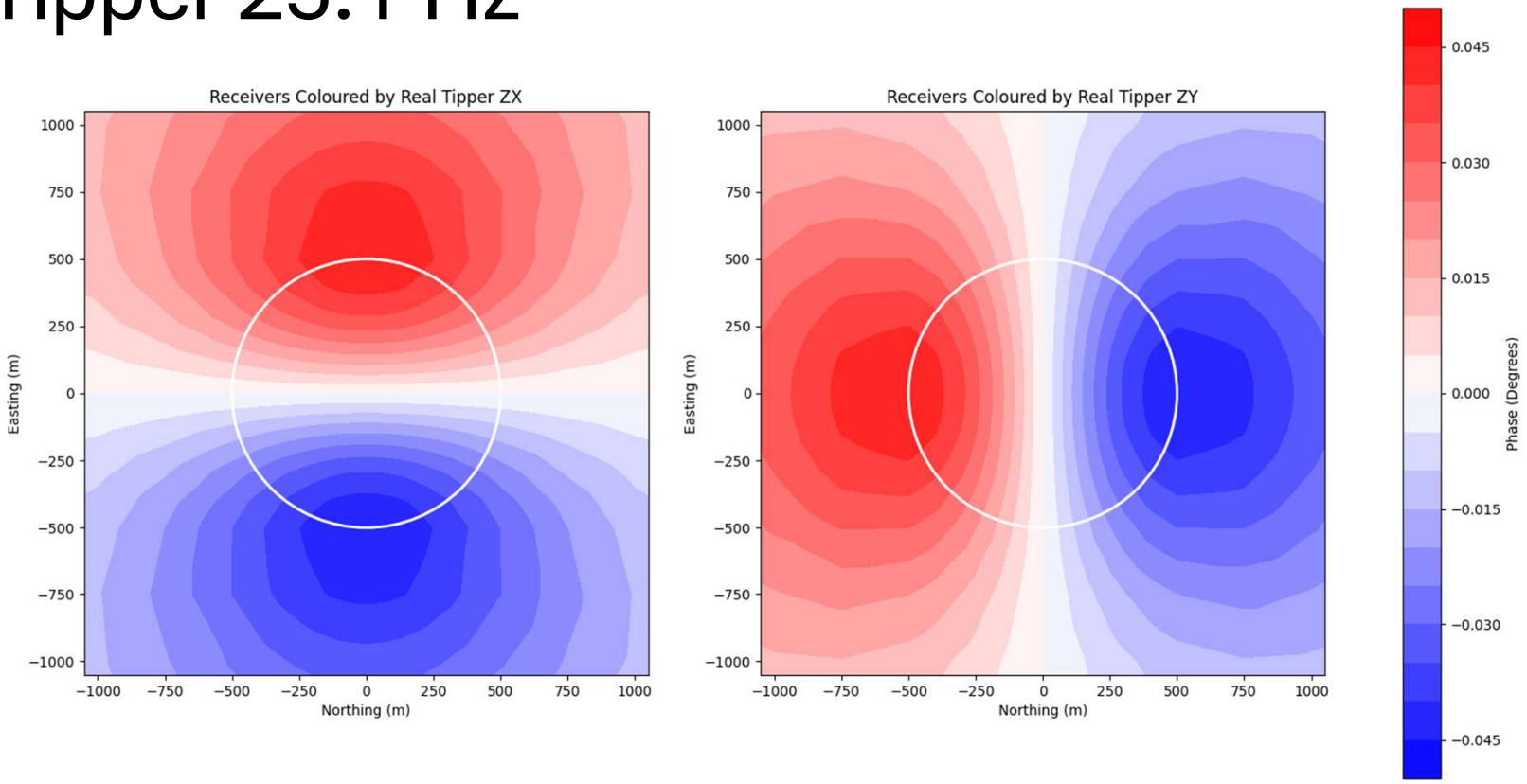
Phase 316 Hz


Receivers Coloured by Phase XY
Receivers Coloured by Phase YX
Easting (m)
Northing (m)
1000
750
500
250
0
−250
−500
−750
−1000
Phase (Degrees)
54.0
52.5
51.0
49.5
48.0
46.5
45.0

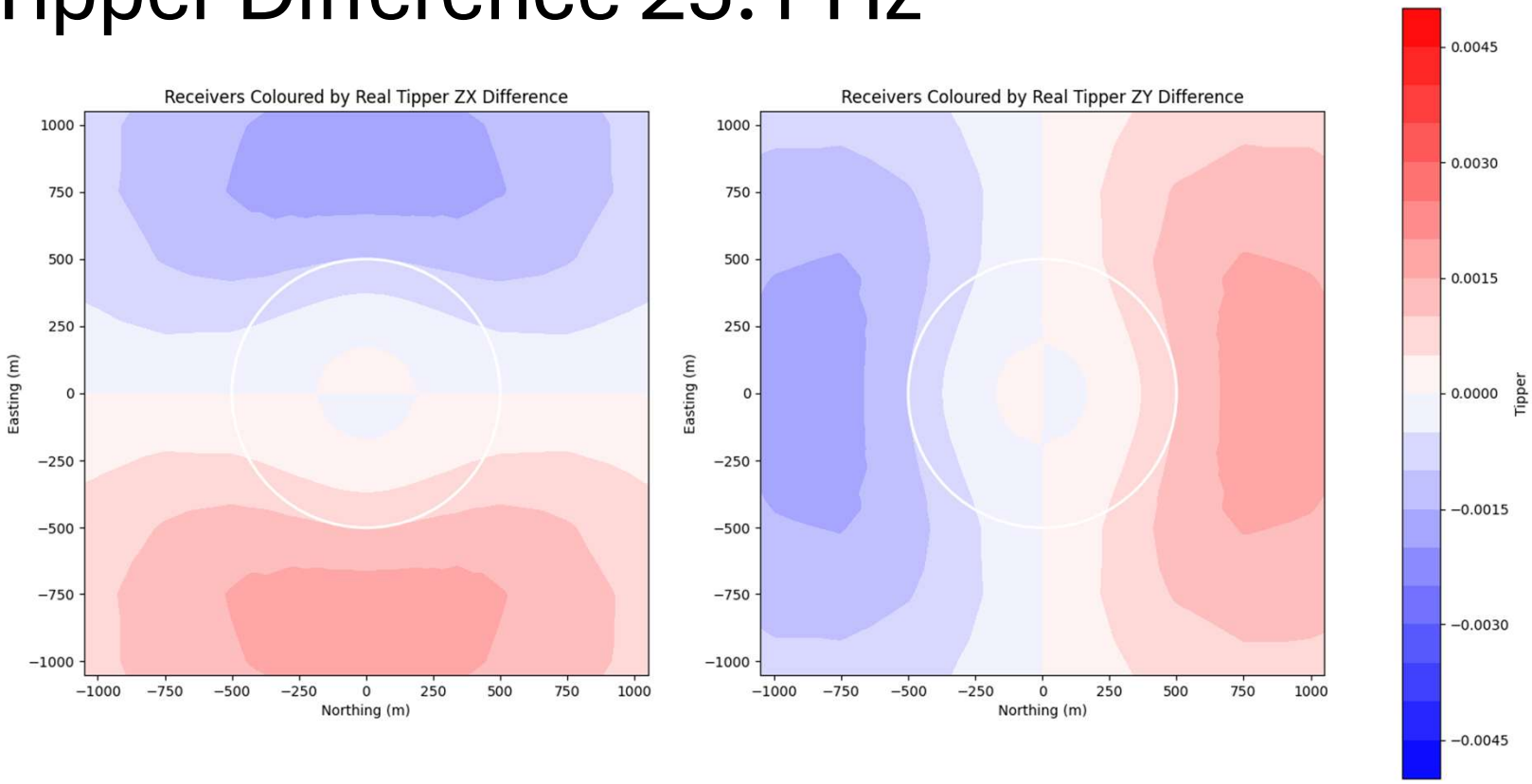
Phase Difference 316 Hz



Tipper 25.1 Hz



Tipper Difference 25.1 Hz



Conclusion

- Some insight was gained into how mesh refinement schemes influence accuracy and computation time
- The effect of volume averaging was studied for the sphere in a halfspace example
- Low frequency residuals are within 1% and all else within 5% in ~4.5 hours simulation time